

Java Theory Assignment

Format: Handwritten (A4 size Paper)

Unit 1: Introduction to Java

1. The Java Philosophy

1. Define Java. Explain the historical context of its development and how it evolved to meet the needs of the World Wide Web.
2. Elaborate on the "Write Once, Run Anywhere" (WORA) principle. How does Java achieve platform independence?

2. Architecture & Execution

1. Explain the role of the Java Virtual Machine (JVM). Why is it often called a "software-based processor"?
2. Distinguish between Source Code, Bytecode, and Machine Code. Why is Bytecode not considered an executable file?
3. Describe the step-by-step process of what happens when you run the commands `javac` and `java` in your terminal.

3. Paradigms & Applications

1. Compare Procedure-Oriented Programming (POP) and Object-Oriented Programming (OOP). Discuss why Java chose the OOP approach for modern software development.
2. What is the fundamental difference between a Java Application and a Java Applet? Discuss why Applets are no longer widely used in modern IT.

4. Errors & Environment

1. List the environment variables required to run Java on a local machine and explain their purposes.
2. Categorize the common errors a programmer might face into Compile-time and Runtime errors. Give two examples of each.

